

Biology - Graduate Level Assessment 7

Instructions

- Answer all questions.

Multiple Choice

Question 1

(2 marks)

What factors affect the rate of transpiration?

- A. Temperature, humidity, and air currents
- B. Plant species, root depth, and wind speed
- C. Soil fertility, irrigation methods, and cloud cover
- D. Water salinity, leaf color, and time of day
- E. Leaf surface area, stem length, and moon phase

Question 2

(2 marks)

What are the different types of point mutations?

- A. Frame shift mutations and nucleotide repeats
- B. Base substitutions and small insertions and deletions
- C. Base additions and large insertions
- D. Silent mutations and missense mutations
- E. Duplication mutations and triplet repeat expansions

Question 3

(2 marks)

Which is a true statement concerning genetic variation?

- A. It is created by the direct action of natural selection.

- B. It tends to be reduced by the processes involved when diploid organisms produce gametes.
- C. It must be present in a population before natural selection can act upon the population.
- D. It is always increased by the processes involved when diploid organisms produce gametes.
- E. It is not affected by the environmental changes.

Question 4

(2 marks)

Which group is composed entirely of individuals who maintained that species are fixed (i.e., unchanging)?

- A. Aristotle, Darwin, and Lamarck
- B. Aristotle, Lyell, and Darwin
- C. Aristotle, Linnaeus, and Cuvier
- D. Aristotle, Cuvier, and Lamarck
- E. Linnaeus, Darwin, and Lamarck

Question 5

(2 marks)

Compare the methods of obtaining oxygen in the earthworm, the frog, the fish and man.

- A. Earthworms use gills, frogs use both gills and lungs, fish use lungs, and humans use skin.
- B. Earthworms and humans use lungs, frogs and fish use gills.
- C. All use gills for obtaining oxygen.
- D. Earthworms and humans use gills, frogs use skin, and fish use lungs.
- E. Earthworms use lungs, frogs use gills, fish use skin, and humans use gills.

True / False

Question 1

(2 marks)

Search engine as a diagnostic tool in difficult immunological and allergologic cases: is Google useful?

True

False

Question 2

(2 marks)

Can increases in the cigarette tax rate be linked to cigarette retail prices?

True

False

Question 3

(2 marks)

Are head and neck specific quality of life measures necessary?

True

False

Question 4

(2 marks)

Did Chile's traffic law reform push police enforcement?

True

False

Question 5

(2 marks)

Is there a correlation between androgens and sexual desire in women?

True

False

Long Form

Question 1*(10 marks)*

Discuss the role of radioactive isotopes in biological research, particularly in tracing compound pathways in living cells, and analyze their impact on cell observation techniques.

Type your answer here...

**Question 2***(10 marks)*

Discuss the role of ethylene in the ripening process of fruits and evaluate the misconceptions surrounding the use of heat from kerosene stoves in this context.

Type your answer here...



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